WORKED SOLUTION

STEP II 1993, Question 12

Mechanics • Friction and Rigid Bodies

STEP II 1993, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q12

The key idea

The sphere is wedged between the wall and the plane, so it can't move off without losing contact with one of them. If it slips at both contacts at once, its centre stays put and it simply spins about its centre. Working out which way it would spin tells us which way each friction force acts. After that it's three equations (two resolutions and one moment equation) in the unknowns N_1, N_2 and m. For the second part, the right moment point makes the answer fall out in one line.

Solution

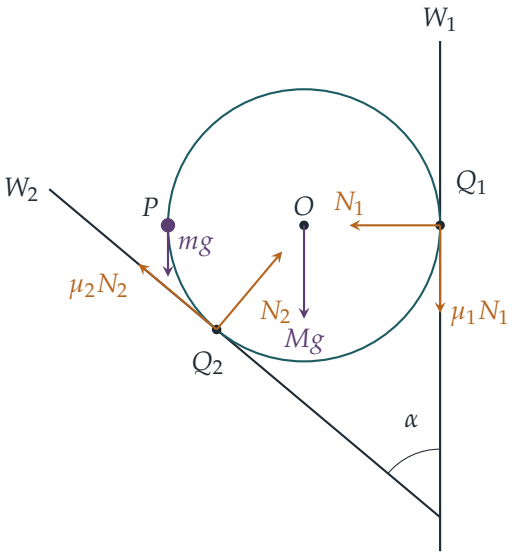
The model and the directions of friction

We model the sphere as a rigid uniform sphere and the particle as a point mass fixed at P. The contacts at Q_1 and Q_2 are rough. Let the normal reactions be N_1 (horizontal, away from the wall) and N_2 (perpendicular to the plane, through the centre O).

Since PQ_1 is a horizontal diameter, the particle hangs at horizontal distance r from O, on the side away from the wall. Its weight therefore has a moment mgr about O, turning the sphere so that P goes down, and the forces at the contacts must balance that.

? **Ask yourself** Which way does friction act at each contact?

If the sphere slips at both points it stays in contact with both, so O is fixed and the sphere rotates about O in the sense in which P falls. Then Q_1, the point opposite P, moves *up* the wall, and Q_2 moves *down* the plane towards the corner. Friction opposes the slipping, so it acts *down* the wall at Q_1 and *up* the plane at Q_2. Since slipping is about to happen at both points, friction is limiting at both: $F_1 = \mu_1 N_1$ and $F_2 = \mu_2 N_2$. Both friction forces have moments about O that oppose the particle's moment, as they should.



Slipping at both contacts

The plane makes angle α with the vertical, so its normal makes angle α with the horizontal. Then N_2 has components $N_2\cos\alpha$ towards the wall and $N_2\sin\alpha$ upwards, and the friction $\mu_2 N_2$ up the plane has components $\mu_2 N_2\sin\alpha$ away from the wall and $\mu_2 N_2\cos\alpha$ upwards.

Resolving horizontally:

$$N_1 = N_2\cos\alpha - \mu_2 N_2\sin\alpha = N_2(\cos\alpha - \mu_2\sin\alpha). \tag{1}$$

Resolving vertically:

$$N_2\sin\alpha + \mu_2 N_2\cos\alpha - \mu_1 N_1 = (M+m)g. \tag{2}$$

Taking moments about O: both normal reactions pass through O, and so does Mg. The two friction forces act along tangents, at distance r from O. So

$$mgr = \mu_1 N_1 r + \mu_2 N_2 r \quad\Longrightarrow\quad mg = \mu_1 N_1 + \mu_2 N_2. \tag{3}$$

Substituting (1) into (3) and into (2):

$$\begin{aligned} mg &= N_2(\mu_2 + \mu_1\cos\alpha - \mu_1\mu_2\sin\alpha),\\ (M+m)g &= N_2(\sin\alpha + \mu_2\cos\alpha - \mu_1\cos\alpha + \mu_1\mu_2\sin\alpha). \end{aligned}$$

Subtracting the first from the second gives

$$\begin{aligned} Mg &= N_2(\sin\alpha + \mu_2\cos\alpha - 2\mu_1\cos\alpha + 2\mu_1\mu_2\sin\alpha - \mu_2)\\ &= N_2((2\mu_1\mu_2+1)\sin\alpha + (\mu_2 - 2\mu_1)\cos\alpha - \mu_2). \end{aligned}$$

Dividing the equation for mg by this one cancels N_2 and g:

$$m = \frac{\mu_2 + \mu_1\cos\alpha - \mu_1\mu_2\sin\alpha}{(2\mu_1\mu_2+1)\sin\alpha + (\mu_2 - 2\mu_1)\cos\alpha - \mu_2}\,M,$$

as required.

For this to make physical sense we need $N_1 > 0$, which by (1) means $\mu_2 < \cot\alpha$. If μ_2 is larger than that, friction at Q_2 would pull the sphere off the wall, and we get the situation in the final part.

Rolling about Q_2

Now the sphere doesn't slip at Q_2. Instead it is about to turn about Q_2 (rolling up the plane), so its centre moves away from the wall and contact at Q_1 is about to be lost. At the point of rolling, $N_1 = 0$, and so the friction at Q_1 is zero too.

? **Ask yourself** What do we do with the friction at Q_2, which is now unknown?

We take moments about Q_2, so that both the reaction and the friction at Q_2 drop out. That leaves only the two weights. Measure horizontal distances from Q_2. Since OQ_2 is perpendicular to the plane, it makes angle α with the horizontal, so O is a horizontal distance $r\cos\alpha$ from Q_2, on the wall side. The particle is a further r beyond O, away from the wall, so it is a horizontal distance $r - r\cos\alpha$ from Q_2, on the other side. For the moments to balance,

$$Mg\,r\cos\alpha = mg\,r(1-\cos\alpha) \quad\Longrightarrow\quad m = \frac{M\cos\alpha}{1-\cos\alpha} = \frac{M}{\sec\alpha - 1},$$

dividing top and bottom by $\cos\alpha$. That is the result.

Checks

- The two answers should agree on the boundary between the two regimes, $N_1 = 0$, that is $\mu_2 = \cot\alpha$. Putting $\mu_2 = \cos\alpha/\sin\alpha$ into the first formula, the numerator becomes $\cot\alpha$ and the denominator becomes $(1-\cos\alpha)/\sin\alpha$, whatever μ_1 is. So $m = M\cos\alpha/(1-\cos\alpha)$, which matches the second answer.
- With $\alpha = 60^\circ$ the second answer gives $m = M$. With $\alpha = 60^\circ$, $\mu_1 = 0.2$, $\mu_2 = 0.3$ and $M = 1$, the first gives $m \approx 0.561$. We checked the force and moment balance for these values, and for 2000 random cases, with a computer.
- If both contacts were smooth ($\mu_1 = \mu_2 = 0$), the first formula gives $m = 0$: nothing can resist the particle's moment, as we'd expect.

Common slips

- Getting a friction direction wrong. Decide how the sphere would turn, then which way each contact point would slide.
- Forgetting that both reactions pass through the centre, so they don't appear in the moment equation about O.

- In the final part, keeping the wall's friction. Once the sphere is about to leave the wall, $N_1 = 0$, so the friction there is zero.

Beyond STEP

Problems like this, with several contacts that may slip or roll, are the start of the mechanics of rigid bodies with constraints. Each contact gives an inequality ($|F| \leq \mu N$, $N \geq 0$), and you have to work out which of them hold with equality. The same case analysis turns up in robotics, where it decides whether a robot hand can grip something. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers statics and the rotation of rigid bodies carefully.

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